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Evgeny Sobolev 09.02.1984 y.b.

$y(x) := \sqrt{x}$

$Max32 := \left(2^{32} - 1\right)$

$clz(x) := \text{if } x = 0 \; 32 \text{ else } 32 - \text{trunc}\left(\log_2\left(\text{trunc}(x) + \frac{Max32}{2^{32}}\right) + \frac{Max32}{2^{32}}\right)$

$msb(x) := 32 - clz(x)$

$dY_32 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{32} - \left(\sqrt{2}\right)^{31}\right) \cdot 2^{16}\right)$

$Y0_32 := \text{trunc}\left(\sqrt{2} \cdot 2^{15} \cdot 2^{16}\right)$

$dY_31 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{31} - \left(\sqrt{2}\right)^{30}\right) \cdot 2^{16}\right)$

$Y0_31 := \text{trunc}\left(2^{15} \cdot 2^{16}\right)$

$dY_32 = 1257966796$

$dY_31 = 889516851$

$Y0_32 = 3037000499$

$Y0_31 = 2147483648$

$dY(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x) \; dY_32 \text{ else } dY_31$

$Y0(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x) \; Y0_32 \text{ else } Y0_31$

$xScaled(x) := \text{trunc}\left(\text{trunc}(x) \cdot 2^{clz(x)+1} - 2^{32}\right)$

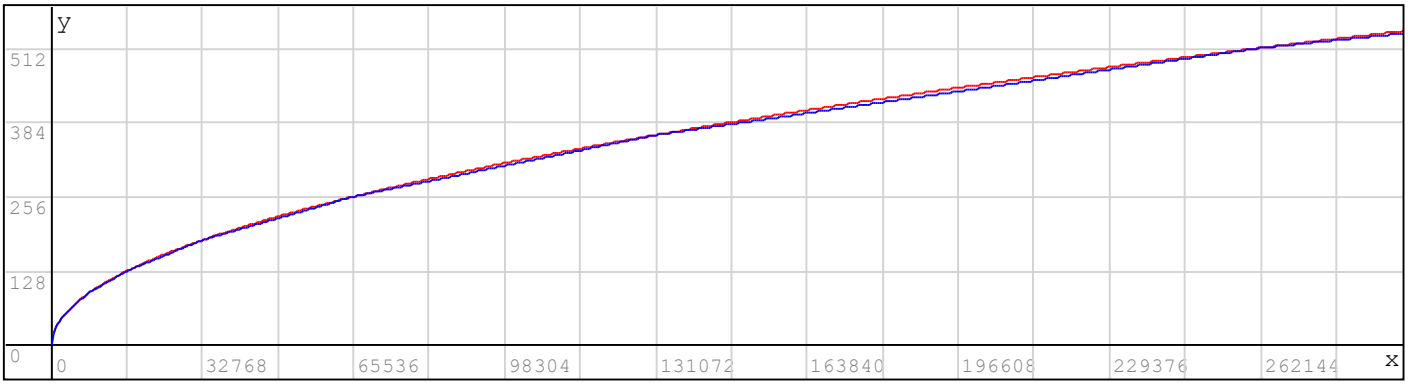
$sqrtLinearScaled(x) := \text{trunc}\left(\frac{\text{trunc}(xScaled(x) \cdot dY(x))}{2^{32}}\right) + Y0(x)$

$sqrtLinear(x) := \frac{sqrtLinearScaled(x)}{2^{\text{trunc}\left(\frac{clz(x)}{2}\right)}}$

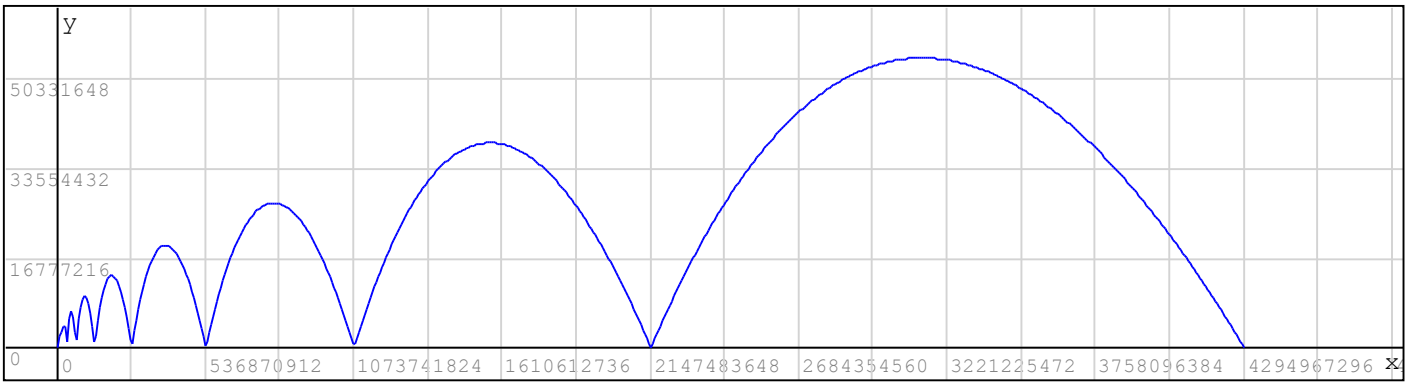
$sqrtLinearErr(x) := y(x) - sqrtLinear(x)$

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$$\left\{ \frac{sqrtLinear(x)}{2^{16}},\ \sqrt{x} \right.$$



$$\left\{ \sqrt{x} \cdot 2^{16} - sqrtLinear(x) \right.$$

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$xMaxErr_32 := \text{trunc}\left(\frac{2^{32}}{\left(4-2\cdot\sqrt{2}\right)^{2}}\right)$ $xMaxErrScaled_32 := \text{trunc}\left(\frac{2\cdot\sqrt{2}-1}{4}\cdot 2^{32}\right)$

$yMaxErr_31 := \text{trunc}\left(\sqrt{\frac{xMaxErr_32}{2}}\cdot 2^{16} - sqrtLinear\left(\frac{xMaxErr_32}{2}\right)\right)$

$yMaxErr_32 := \text{trunc}\left(\sqrt{xMaxErr_32}\cdot 2^{16} - sqrtLinear\left(xMaxErr_32\right)\right)$

$xMaxErr_32 = 3129112985$ $yMaxErr_31 = 38154242$ $yMaxErr_32 = 53958246$

$xMaxErrScaled_32 = 1963258675$

$yMaxErrScaled(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right)\cdot 2 = clz(x)$ $yMaxErr_32$ else $yMaxErr_31$

$\frac{yMaxErr_32}{65536} = 823.33749$

$xMaxErr(x) := \frac{xMaxErr_32}{2^{\text{trunc}(clz(x))}}$ $yMaxErr(x) := \frac{yMaxErrScaled(x)}{2^{\text{trunc}\left(\frac{clz(x)}{2}\right)}}$

$dXMaxErr(x) := \left|\left(xScaled(x) - xMaxErrScaled_32\right)\cdot 2\right|$

$$dXMaxErr(x) := \text{if } xScaled(x) < xMaxErrScaled_32 \left|\left(xMaxErrScaled_32 - xScaled(x)\right)\right|\cdot\frac{Max32}{xMaxErrScaled_32} \text{ else } \left|\left(xScaled(x) - xMaxErrScaled_32\right)\right|\cdot\frac{Max32}{Max32 - xMaxErrScaled_32}$$

$dQMaxErr(x) := \text{trunc}\left(\frac{\left(dXMaxErr(x)\right)^{2}}{2^{32}}\right)$ $dXMidErr(x) := \left|2^{31} - xScaled(x)\right|\cdot 2$

$dQMaxErrInv(x) := 2^{32} - dQMaxErr(x)$ $dQMidErr(x) := \text{trunc}\left(\frac{\left(dXMidErr(x)\right)^{2}}{2^{32}}\right)$

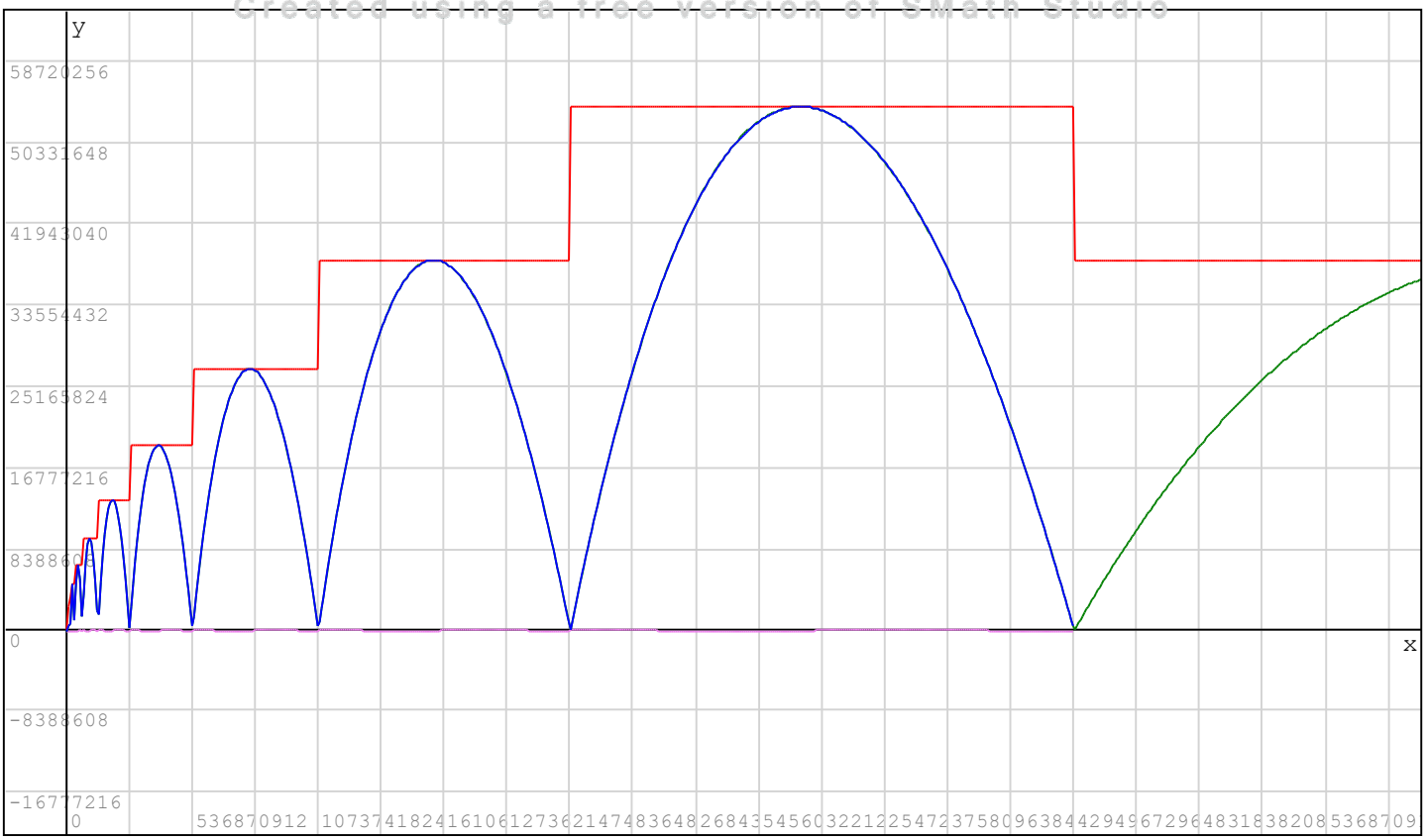
$parMaxErr(x) := \text{trunc}\left(\frac{dQMaxErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right)$ $dQMidErrInv(x) := 2^{32} - dQMidErr(x)$

$parMidErr(x) := \text{trunc}\left(\frac{dQMidErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right)$

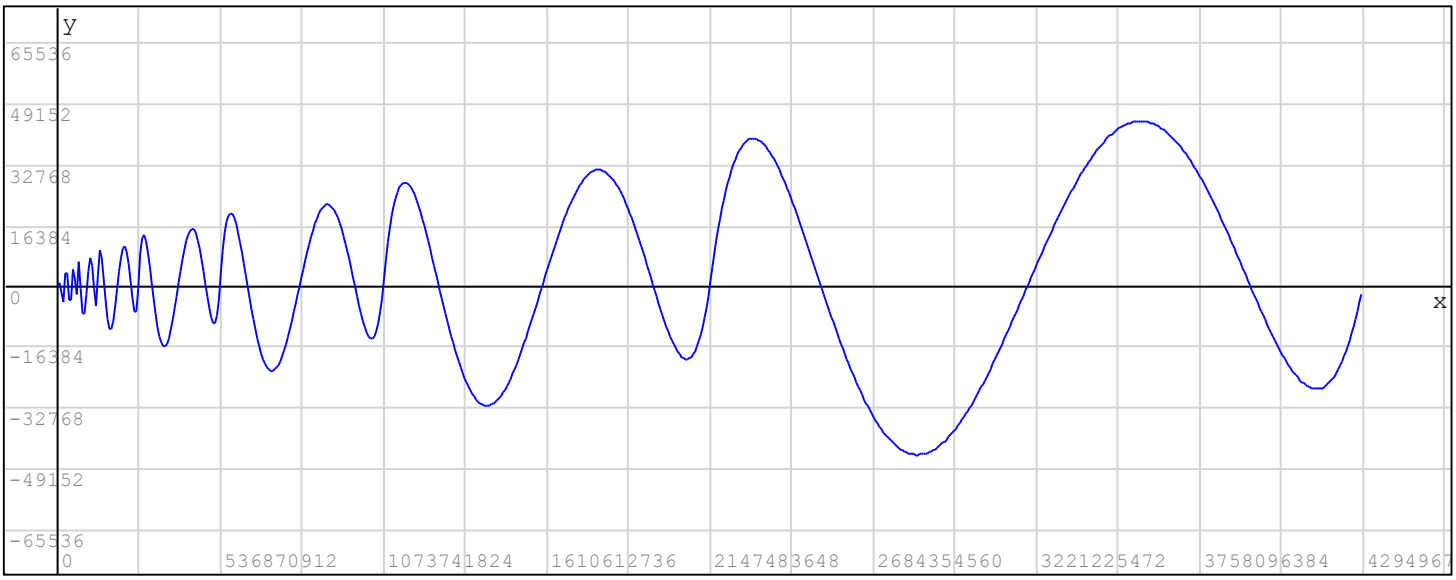
$par(x) := \left(\text{trunc}\left(\frac{parMaxErr(x)\cdot dQMidErrInv(x)}{2^{32}}\right) + \text{trunc}\left(\frac{parMidErr(x)\cdot dQMidErr(x)}{2^{32}}\cdot(1.0365)\right)\right)$

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$$\begin{cases} \sqrt{x}\cdot 2^{16} - sqrtLinear(x) \\ yMaxErr(x) \\ \dfrac{par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}} \\ \sqrt{x}\cdot 2^{16} - \dfrac{sqrtLinearScaled(x)+par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}} \end{cases}$$



$$\begin{cases} \sqrt{x}\cdot 2^{16} - \dfrac{sqrtLinearScaled(x)+par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}} \end{cases}$$

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$$parErr(x) := \sqrt{x} \cdot 2^{16} - \frac{sqrtLinearScaled(x) + par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}}$$

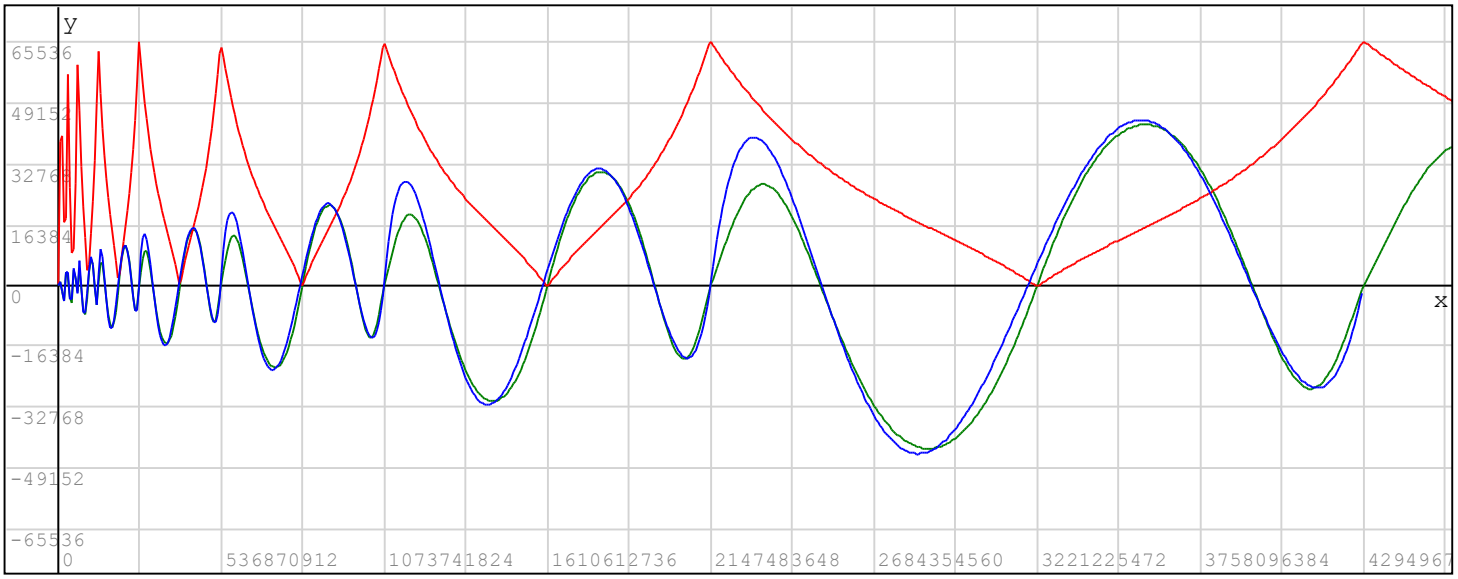
$A := 0.092 \quad B := 0.95 \quad C := 0.00$

$Aamp := 0.695$

$$parArgCorr(x) := \left(\frac{dXMidErr(x)}{2^{32}} \cdot (1 - A - C)\right) + \frac{dQMidErr(x)}{2^{32}} \cdot (A - B) + \frac{dQMidErr(x) \cdot dXMidErr(x)}{2^{64}} \cdot B + C$$

$$Amp(x) := \frac{dXMidErr(x)}{2^{32}} \cdot (1 - Aamp) + \frac{dQMidErrInv(x)}{2^{32}} \cdot Aamp$$

$$parCorr(x) := \sin\left(\operatorname{sign}\left(xScaled(x) - 2^{31}\right) \cdot (parArgCorr(x) \cdot \pi \cdot 2)\right) \cdot Amp(x) \cdot 44544 \cdot 1.35$$



$$\begin{cases} parErr(x) \\ parArgCorr(x) \cdot 65536 \\ \dfrac{parCorr(x)}{2^{\frac{clz(x)}{2}}} \end{cases}$$

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